Midterm 2 Grading Rubric Notes

- **#3** -> 4 points for each answer: 2 points for listing a non-coding region (e.g. promoter, intron) and then 2 more points for correctly describing how a mutation would reduce levels. If some type of answer is given, but that would cause increase levels, 1 point. If the description makes no sense, 2 points are not given

- **#7-8** -> If #7 is wrong, but selected the correct corresponding ratios in question #8, then full credit is given for #8 (#7 would still be marked as wrong)

- **#11** -> if the term "suppression" or "genetic suppressor" is provided, full credit is given. Some partial credit is given in certain cases. Mentioning epistasis, is not correct.

Point adjustment

Question 7 – 4 points, 23% correct
Question 9 – 4 points, 15% correct
Question 11 – 4 points, 27% correct
Question 13 – 4 points for F1 genotype/phenotype, ~31% correct

Question 12 (24 points total)

Allele	Cases	Controls	
G	$\mathbf{O_{G_case}}$ = 5,100 $\mathbf{E_{G_case}}$ = (Total G / Total genomes) x Total Cases = (7,456/14,872) * 8,415 = 4218.82 $\mathbf{X^2_{G_case}}$ = $(E_{G_case} - O_{G_case})^2 / E_{G_case}$ = (4218.82 - 5100)^2 / 4218.82 = 184.05	$\mathbf{O_{G_ctrl}}$ = 2,356 $\mathbf{E_{G_ctrl}}$ = (Total G/Total genomes) x Total Controls = (7,456/14,872)*6,457 = 3237.18 $\mathbf{X^2_{G_ctrl}}$ = $(E_{G_ctrl} - O_{G_ctrl})^2 / E_{G_ctrl}$ = (3237.18 - 2356)^2 / 3237.18 = 239.86	Total G = 7,456
A	$\mathbf{O_{A_case}}$ = 3,315 $\mathbf{E_{A_case}}$= (Total A/Total genomes) x Total Cases = = (7,416/14,872) * 8,415 = 4196.18 $\mathbf{X^2_{A_case}}$ = $(E_{A_case} - O_{A_case})^2 / E_{A_case}$ = (4196.18 - 3315)^2 / 4196.18 = 185.04	$\mathbf{O_{A_ctrl}}$ = 4,101 $\mathbf{E_{A_ctrl}}$ = (Total A/Total genomes) x Total Controls = (7,416/14,872) * 6,457 = 3219.82 $\mathbf{X^2_{A_ctrl}}$ = $(E_{A_ctrl} - O_{A_ctrl})^2 / E_{A_ctrl}$ = (3219.82 - 4,101)^2 / 3219.82 = 241.16	Total A = 7,416
	Total Cases = 8,415	Total Controls = 6,457	Total genomes = 14,872

(6 points) χ^2 statistic:

$$\chi^2 = \sum(Oi - Ei)^2 / Ei$$
$$= \mathbf{X^2_{G_case} + X^2_{G_ctrl} + X^2_{A_case} + X^2_{A_ctrl}}$$
$$= 184.05 + 239.86 + 185.04 + 241.16$$
$$= \mathbf{850.11}$$

<u>Points breakdown:</u>

Correct test statistic: 1 point

Showed work: 5 points

(6 points) Degrees of freedom:

There is **one degree of freedom** as there are two categories (and each individual is in exactly one category, A or G) so there is one "free" component in the random variable.

<u>Points breakdown:</u>

Correct degrees of freedom: 6 points

(6 points) *p*-value:

The test statistic is very large (for a chi-squared value) and corresponds to an extremely small p-value, so many lookup tables/programs will fail[1]. Full credit for "zero" or similar.

***p*-value: 6.88×10^{-187}**
found with python: `scipy.stats.chi2.sf(850.11, 1)`

<u>Points breakdown:</u>

Correct *p*-value (zero, 6.88×10^{-187} or similar): 6 points
Incorrect *p*-value but showed work: up to 4 points

(6 points) Significance

We are considering 500,000 SNPs, so we need to correct for multiple testing.

Our *p*-value is **significant** because it is less than **0.05 / 50,0000 = 1×10^{-7}**

<u>Points breakdown:</u>

Correct significance: 1 point
Performed/attempted multiple testing: up to 5 points

[1] Most programs get the p-value by evaluating the CDF of the chi-squared distribution at a point *y*, which gives $P[X \leq y]$, and a p-value of $1 - P[X \leq y]$ (probability that a random variable $X \sim \chi^2$ is greater than your test statistic). For large/small values precision becomes an issue for computers.

#13 – Answer Key

① BBPP × bbpp ⟶ $\underbrace{BbPp}_{\text{genotype}}$ = $\underbrace{\text{all black}}_{\text{phenotype}}$

+4 for correct genotype

+4 for correct phenotype

② BbPp × BbPp

	BP	Bp	bP	bp
BP	BBPP	BBPp	BbPP	BbPp
Bp	BBPp	BBpp	BbPp	Bbpp
bP	BbPP	BbPp	bbPP	bbPp
bp	BbPp	Bbpp	bbPp	bbpp

9 black

4 albino

3 cream

+2 for showing work/ genotypes

+6 for correct phenotypic ratio based on #1

③ Recessive epistasis

(homozygous recessive pp for albinism hides the expression of all B/b alleles)

+8 for "recessive epistasis" or "epistasis" with explanation

(+6 for just "epistasis")